



# Edema

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## Introduction

Edema indicates an underlying condition that requires evaluation and diagnosis, or it may be drug induced. The underlying disorder should be diagnosed and treated concurrently with initiation of directed therapy for the edema. Edema is caused by an imbalance of fluid homeostasis that leads to increased extracellular fluid volume (ECFV). Specifically, it reflects an increase in the major cation (sodium) and anion (chloride) of the ECFV.

Peripheral edema is swelling of a dependent body part (usually legs and sometimes arms) that is visible on physical examination. Palpation can differentiate pitting (indentation left when pressure applied) from nonpitting edema. Anasarca refers to clearly evident, generalized edema. To be visible on the physical exam, the ECFV is increased by many litres. It takes many days for edema to develop when the intake of NaCl exceeds the excretion of NaCl.

Common conditions associated with edema include heart failure (HF); liver failure, including hepatic cirrhosis; and kidney failure, especially nephrotic syndrome. The main focus of this chapter is on the management of peripheral edema. For edema associated with specific conditions, see [Heart Failure](#) and [Chronic Liver Diseases](#).

## Goals of Therapy

- Decrease fluid volume overload
- Improve patient comfort and quality of life
- Identify and manage diuretic resistance

## Investigations

- History:
  - conditions associated with edema:
    - general conditions: kidney dysfunction, HF, liver dysfunction, pregnancy, low serum albumin, capillary leak, hypothyroidism (myxedema)
    - localized conditions: lymphatic or venous obstruction (e.g., due to malignancy), infection or inflammation (e.g., bee sting or snake bite)
  - salt intake
  - drugs that may cause or exacerbate edema (see [Table 1](#))
- Physical exam to assess total body sodium (reflects ECFV):
  - generalized edema
  - localized edema:
    - pitting or nonpitting—for nonpitting edema, consider lymphatic obstruction or myxedema
    - symmetric or asymmetric—for asymmetric edema, consider lymphedema from lymphatic obstruction, deep vein thrombosis (DVT), chronic venous insufficiency, infection or sequelae of prior surgery on a distal lower (such as saphenous venous graft donation for coronary artery bypass grafting [CABG]) or upper (such as prior arteriovenous fistula/graft) extremity
  - site of edema:
    - diffuse edema in anasarca, especially periorbital edema
    - dependent edema: pedal edema, legs if patient ambulatory, sacrum if patient recumbent, scrotum
    - compartmentalized: ascites, pleural effusion, pericardial effusion
  - weight, BP (hypertension is a feature of expanded ECFV; hypotension, especially postural changes, reflects contracted ECFV), heart rate, respiratory rate, skin turgor or shine, assessment of abdomen size, abdominal percussion (if ascites present), or chest percussion (if pleural effusion present)
  - elevation of the jugular venous pressure (JVP), ascites and pleural effusions are features of expansion of the ECFV; loss of skin turgor and decreased JVP are features of contraction of the ECFV
- Laboratory investigations:
  - serum electrolytes (including Na<sup>+</sup>, K<sup>+</sup>, Cl<sup>-</sup>, HCO<sub>3</sub><sup>-</sup>, Ca<sup>++</sup>, Mg<sup>++</sup>), uric acid, urea, creatinine
  - urinalysis to detect excessive albumin in the urine, which is seen in nephrotic syndrome
  - liver function tests, especially those that reflect synthetic dysfunction of the hepatocytes (INR, albumin, ammonia)

- other: TSH, blood glucose and lipids, natriuretic peptide (BNP or NT-proBNP) to screen for heart failure, serum albumin (low in liver disease or proteinuric kidney disease), 24-hour urine collection to quantify proteinuria and sodium if symptoms/signs are suggestive of primary or secondary kidney disorder
- Diagnostic imaging:
  - chest x-ray to determine cardiac size and assess lung fields
  - point-of-care ultrasound (POCUS). Lung POCUS can assess interstitial edema and pleural effusions. Cardiac POCUS can assess pericardial effusion. Central venous pressure can be assessed by imaging blood content of the vena cavae

**Table 1: Examples of Drugs that Can Cause Edema**<sup>[1]</sup>

| Medication Class          | Drug                                                       |
|---------------------------|------------------------------------------------------------|
| Antidepressants           | escitalopram, mirtazapine, paroxetine, venlafaxine         |
| Antiepileptics            | carbamazepine, clobazam, valproic acid                     |
| Antihypertensives         | alpha-1 blockers, beta-blockers, CCBs, minoxidil,          |
| Antipsychotics            | clozapine, olanzapine, quetiapine, risperidone             |
| Antispastics              | baclofen                                                   |
| Calcineurin inhibitors    | cyclosporine, tacrolimus                                   |
| Checkpoint inhibitors     | nivolumab                                                  |
| Corticosteroids           | prednisone                                                 |
| Estrogens                 | estradiol                                                  |
| NSAIDs                    | celecoxib, naproxen                                        |
| Opioids                   | hydromorphone, morphine, oxycodone, tramadol               |
| Progestins                | progesterone                                               |
| Protein kinase inhibitors | BCR-ABL: bosutinib, dasatinib, imatinib<br>VEGF: sunitinib |
| Proton pump inhibitors    | lansoprazole, omeprazole, pantoprazole                     |
| SERMs                     | tamoxifen                                                  |

**Abbreviations:** BCR-ABL = breakpoint cluster region-Abelson; CCB = calcium channel blocker; NSAID = nonsteroidal anti-inflammatory drug; SERM = selective estrogen receptor modulator; VEGF = vascular growth factor

## Therapeutic Choices

[Figure 1](#) suggests a strategy for the management of edema.

### Nonpharmacologic Choices

- Sodium: decrease sodium intake to <88 mmol/day (2 g Na<sup>+</sup> or 5 g NaCl)<sup>[2][3]</sup>
- Fluid: decrease free-water intake based on individual patient assessment of volume and disease state, especially if hyponatremia is also present (<130 mmol/L)
- Posture: supine position augments venous return to the heart, which can exacerbate HF symptoms in a patient with decompensated HF (orthopnea), especially in bed at night (paroxysmal nocturnal dyspnea), so it should be avoided in these patients
- Stockings: consider supportive compression stockings (medium pressure compression stocking: 20–30mm/Hg); too much compression can be detrimental in severe peripheral vascular disease
- Paracentesis for ascites: performed under the direction of a specialist<sup>[3]</sup>
- Ultrafiltration therapy: if kidney insufficiency is significant, consider referral to a specialist for this therapy

### Pharmacologic Choices

Consider if nonpharmacologic actions do not significantly reduce the symptoms related to the edema and if treatment of the underlying disorder has been implemented.

### Diuretic Therapy

Diuretics (see [Table 2](#)) decrease sodium reabsorption at various sites in the nephron by interfering with NaCl reabsorption via varied solute transporters, channels and/or pumps.<sup>[4][5]</sup> Diuretics increase sodium chloride and typically free water losses; this decreases extracellular fluid volume and improves the edema. Closely monitor for electrolyte imbalances (K<sup>+</sup>, Cl<sup>-</sup>, Ca<sup>++</sup>, Mg<sup>++</sup>, HCO<sub>3</sub><sup>-</sup>) and prekidney azotemia (urea, creatinine) to identify and respond to significant alterations.

Peripheral edema, in the absence of respiratory or cardiac compromise (shortness of breath, decreased functional capacity, chest pain), is not life-threatening, and fluid removal must be slow to avoid intravascular volume depletion relative to interstitial fluid. Accurate daily weights are a priority. The maximum recommended rate of weight loss is 1 kg/day if significant peripheral edema is present.<sup>[3][4][6]</sup> Diuretics are not effective if daily intake of NaCl is not limited. Diuretics are also not effective when

lymphatic or venous drainage obstruction is the underlying problem. Diuretics should not be prescribed in pregnancy (see [Choices during Pregnancy and Breastfeeding](#)).

### Loop Diuretics

The loop diuretics **furosemide**, **ethacrynic acid**, **bumetanide** and **torsemide** (not available in Canada) are potent diuretic agents that act in the loop of Henle in the nephron, specifically inhibiting the Na-K-2Cl transporter located at the luminal membrane of the medullary thick ascending limb. The onset and duration of action are short. Loop diuretics are particularly useful in the presence of moderate to severe kidney dysfunction (CICr  $\leq$  50 mL/min). If there is no response with the maximum dose, switching to an alternative loop diuretic is unlikely to improve efficacy.<sup>[7]</sup>

Bumetanide has no clear advantage over furosemide despite high bioavailability and different sites of metabolism. Oral and IV doses of bumetanide are nearly equivalent, but the IV formulation is not available in Canada. The oral bioavailability of furosemide in healthy individuals is approximately 50%. Ethacrynic acid is reserved for use in patients with known hypersensitivity to sulfonamide diuretics. There are reports of ototoxicity with all the loop diuretics including ethacrynic acid.<sup>[4][5][6]</sup> Furosemide-induced ototoxicity can cause temporary (<24 h) or permanent hearing loss. This risk is highest with high-dose IV therapy (especially when dosing exceeds 240 mg/h) or at lower doses in patients with impaired kidney function or those taking other ototoxic medications, such as aminoglycosides. Ethacrynic acid may have a greater ototoxic effect at high doses compared to furosemide.

### Potassium-Sparing Diuretics

**Spironolactone** and **eplerenone** are steroidal aldosterone receptor antagonists and therefore may be helpful in situations where aldosterone levels are raised. Spironolactone is recommended in patients with hepatic disease, either alone or in combination with furosemide.<sup>[8]</sup> Spironolactone or eplerenone is recommended as part of therapy for HF.<sup>[9]</sup> Eplerenone is a more selective antagonist for aldosterone with a low affinity for progesterone and androgen receptors. This may result in lower propensity for side effects such as gynecomastia and erectile dysfunction<sup>[10]</sup> (see also [Heart Failure](#) and [Chronic Liver Diseases](#)). **Finerenone** is a nonsteroidal aldosterone receptor antagonist that may be useful in CKD patients with HF, especially in diabetic patients.

**Amiloride** and **triamterene** are potassium-sparing diuretics that act independently of aldosterone in the collecting duct (direct inhibitors). They are useful in combination therapy with loop diuretics for resistant edema or if hypokalemia complicates the initial diuretic therapy used.<sup>[5]</sup>

### Thiazides

Thiazide diuretics are useful first-line agents if the creatinine clearance is >50 mL/min.<sup>[5]</sup> Thiazides work in the distal convoluted tube of the nephron and have a longer duration of action than the loop diuretics. Their specific site of action is the NaCl cotransporter at the luminal membrane of the distal convoluted tubule. They are effective antihypertensive agents and are useful adjuncts to loop diuretic therapy for resistant edema. Hyponatremia, as a complication of diuretic therapy, is more common with thiazide diuretics,

perhaps mediated by total body solute depletion.<sup>[3]</sup> Alternate thiazides, such as **chlorthalidone**, may be more potent and act for a longer duration, but have no major clinical advantage over **hydrochlorothiazide** given in adequate doses in patients with normal kidney function.<sup>[5][11]</sup>

### Vasopressin V<sub>2</sub>-Receptor Antagonists

**Tolvaptan** is the only selective vasopressin V<sub>2</sub>-receptor antagonist (vaptan) available in Canada. Antagonism of vasopressin causes an increase in urine water excretion resulting in increased free water clearance. Use of vaptans should be limited to directed specialist care in cases of volume overload with significant hyponatremia when other measures fail, including the use of urea to augment free water excretion.

### Management of Resistant Edema

- Sodium restriction
  - emphasize adherence to  $\leq$ 88 mmol/day (2 g Na<sup>+</sup> or 5 g NaCl)
  - assess adherence to sodium chloride restriction with 24-hour urine sodium assessment
- Fluid restriction (especially with hyponatremia)
  - emphasize adherence to recommended fluid intake, which may be  $\leq$ 1 L/day if hyponatremia significant
- Bed rest
  - advise supine position with elevated legs (increases glomerular filtration rate)
- Adherence to prescribed diuretic regimen (dose and frequency)
- Dosing of furosemide
  - increase the dose incrementally until diuresis is achieved by doubling the dose every 3–5 days
  - recognize that secretion of the diuretic into the tubular lumen of the nephron can be impaired in severe hypoalbuminemia (albumin <20 g/L) or decreased kidney perfusion states such as kidney failure or HF, resulting in ineffective diuresis with initial doses
  - increase the frequency of dosing to 2–3 times per day if unable to maintain an acceptable duration of effective diuresis
  - double the dose if switching from IV to oral route
  - consider IV dosing if the GI tract is edematous or in decreased cardiac output states that can lead to impaired absorption of oral drug
  - use continuous IV infusion if appropriate, as it may offer some advantage in the critical care setting
  - administer larger doses of loop diuretics in patients with a reduced glomerular filtration rate (GFR) to achieve the desired effect
  - recognize that a point is reached where further increases in dosing or plasma concentration result in minimal additional diuretic effect; this is known as the maximum effective dose
  - monitor clinical status carefully, especially weight, to ensure daily weight loss does not exceed 1 kg

- monitor for and manage electrolyte disturbances, especially hypokalemia and metabolic alkalosis, and pre-kidney azotemia (elevated urea relative to creatinine)

### Combination Diuretic Therapy

- Adding a **thiazide diuretic**, such as metolazone, to a **loop diuretic** adds blockade at a distal site of sodium reabsorption to augment diuresis. If the thiazide is given orally with IV furosemide, administer 30–60 minutes prior to the loop diuretic. If both diuretics are taken orally, administer at the same time.
- **Potassium-sparing diuretics** can be added to further block sodium reabsorption, particularly if the patient is hypokalemic. As life-threatening hyperkalemia can result, exercise caution and monitor electrolytes frequently if the patient has kidney failure, diabetes mellitus or is taking any of the following medications: ACE inhibitors, angiotensin receptor blockers, beta-blockers, cyclosporine, NSAIDs, pentamidine, tacrolimus or trimethoprim.
- **Aldosterone antagonists** (both steroidal and nonsteroidal types) are effective adjunctive agents in the presence of activation of the renin/angiotensin/aldosterone (RAS) pathway, e.g., hepatic disease, HF or nephrotic syndrome.
- **Carbonic anhydrase inhibitors** such as **acetazolamide** may occasionally be added on the advice of a specialist if diuresis is ineffective using a combination of other diuretic agents or if diuresis is complicated by metabolic alkalosis. As life-threatening metabolic acidosis or hypokalemia can result, exercise caution and monitor serum bicarbonate and potassium frequently.
- If all above measures fail, a specialist may consider other options such as ultrafiltration in kidney failure or paracentesis in cirrhosis.

### Idiopathic Edema

- This condition occurs most often in postpubertal, young to middle-aged females with obesity. (*Note: Studies presenting this finding classify patients as male or female. CPhA is aware these terms may not reflect the identity of the individual patient presenting and are not inclusive of all patients.*) It is a diagnosis of exclusion when physical exam and laboratory investigations fail to define an underlying cause (cardiac, hepatic or kidney) for edema. In 80% of cases, the edema is orthostatic; that is, the edema develops in the standing position and is more evident at the end of the day. The edema usually presents in the lower extremities but can also present in the face, hands and breasts. This can be demonstrated by weight gains of 0.7–1 kg/day from morning to evening weight.<sup>[12]</sup> An altered homeostatic response to an upright position promotes the retention of sodium and water. Cyclical increases and decreases in dietary sodium intake and output (possibly associated with unreported diuretic usage) may be implicated in select patients.
- Consider nonpharmacologic measures such as restriction of sodium and fluid intake, recumbent position, avoidance of prolonged periods of standing and use of supportive compression stockings.
  - Consider discontinuation of diuretic therapy (stopping and starting diuretics can precipitate cyclical gains and losses of extracellular fluid, which is sometimes called cyclical edema).
  - Reassess fluid-restricted diet if rebound edema persists when diuretics are stopped. Sodium retention due to neurohormonal adaptation may persist for 1–3 weeks before spontaneous induction of diuresis.
  - Consider intermittent daily diuretic use only when nonpharmacologic measures fail.

- The usage of other therapies (e.g., dopamine agonists, ephedrine or metformin) should be overseen by specialists.<sup>[13]</sup>

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## Choices during Pregnancy and Breastfeeding

Patients who are pregnant and have edema should be referred to a specialist in hypertensive disorders of pregnancy. Diuretics are usually not recommended in order to avoid intravascular contraction and potential negative effects on placental perfusion. Impairment in placental perfusion and blood flow insufficiency have been implicated in the etiology of gestational preeclampsia.

Edema is usually self-limiting in postpartum patients. For breastfeeding patients, **hydrochlorothiazide** and **spironolactone** can be used, since the transfer of these medications into breast milk is very low and no adverse effects have been documented in infants. However, intense diuresis should be avoided as, in theory, diuretics may compromise milk production by depleting intravascular volume. Consider referral to a specialist.

A discussion of general principles on the use of medications in these special populations can be found in [Drugs Use during Pregnancy](#) and [Drug Use during Breastfeeding](#). Other specialized reference sources are also provided in these appendices.

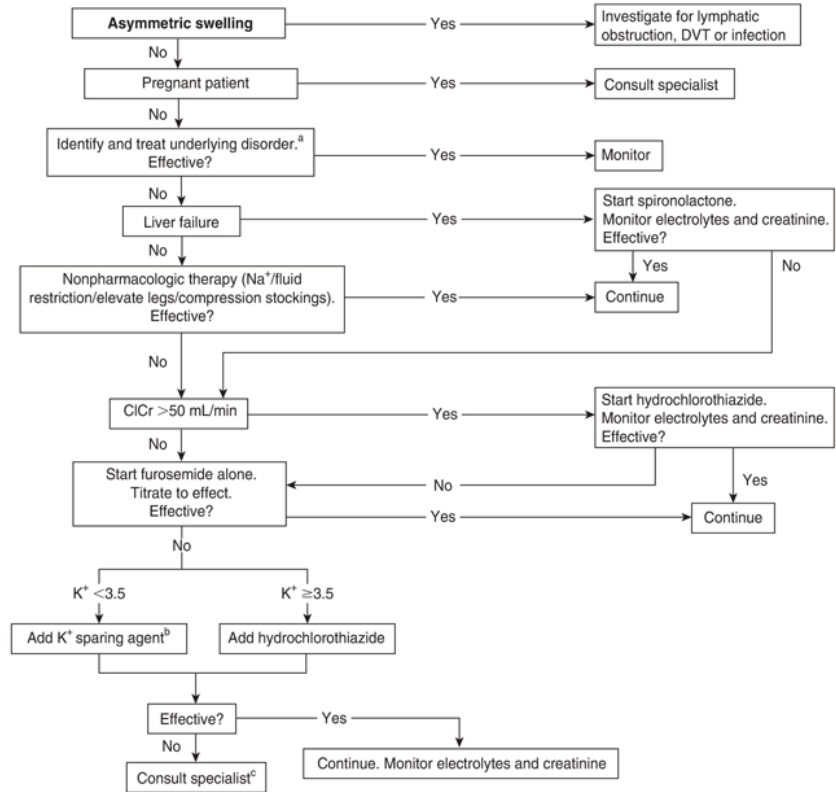
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## Therapeutic Tips

- Investigate unilateral edema for venous obstruction or infection.
  - Monitor daily weight, electrolytes and kidney function closely when initiating diuretic therapy, increasing the dose or using combination therapy, and/or when the patient is clinically unwell.
  - Avoid using potassium-sparing agents for patients with kidney failure; if they are used, follow potassium levels closely as these agents may cause life-threatening hyperkalemia.
  - Avoid using fixed-combination products, as dosing is more difficult to titrate or decrease if changes in volume status occur.
  - Identify overdiuresis in patients by monitoring for clinical signs of hypoperfusion (confusion, dizziness, fatigue, hypotension or weakness) and/or laboratory evidence of prekidney azotemia, hypokalemia and/or metabolic alkalosis. Consider holding diuretics and antihypertensives, using gentle hydration, and/or referral of these patients to a specialist.
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Algorithms

Figure 1: Management of Edema



[a] Examples of disorders to be treated include DVT, heart failure, kidney failure, lymphedema.  
[b] K<sup>+</sup> sparing agents include rapidly acting (e.g., amiloride or triamterene) or long-acting (e.g., spironolactone, eplerenone) drugs.  
[c] Consult kidney or heart or liver specialist as appropriate.

Abbreviations: DVT = deep vein thrombosis

Drug Table

Table 2: Diuretic Agents for Treatment of Edema

| Drug/Cost <sup>[a]</sup>                                                                                                    | Dosage                                                                                                                                                                                                                                                                                                                         | Adverse Effects                                                                                                                                                                                                                                                                                                                                                                                          | Drug Interactions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Comments                                                                                                                                                                                                                                                                                                                                                                                                                       |
|-----------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Drug Class: Loop Diuretics                                                                                                  |                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>bumetanide</b> <br>Burinex<br>\$60–80 | 0.5–2 mg PO as single dose; maximum daily dose 10 mg<br><br>If diuretic response not adequate with 1 mg, a second and third dose may be given at 4- to 5-h intervals<br><br>May give BID<br><br>Maintenance: lowest effective daily dose<br><br>Impaired hepatic function: use minimum effective dose; maximum daily dose 5 mg | Volume depletion.<br>Electrolyte disturbances: hypochloremia, hypokalemia (evident during first 1–2 wk), hyponatremia, hyperuricemia (usually asymptomatic).<br><br>Azotemia, increase in serum creatinine.<br><br>Hyperglycemia.<br><br>Dizziness, hypotension.<br><br>Acute interstitial nephritis.<br><br>Ototoxicity.<br><br>Myalgias (common with high doses of bumetanide).<br><br>Allergy (rare). | Aminoglycosides, cisplatin: increased ototoxicity.<br>Thiazide diuretics: increased diuresis and electrolyte abnormalities; monitor for dehydration and electrolyte abnormalities when used in combination.<br><br>Digoxin: hypokalemia and hypomagnesemia may lead to arrhythmias.<br><br>Lithium: reduced kidney clearance of lithium resulting in lithium toxicity; monitor lithium levels and adjust dose as needed.<br><br><a href="#">NSAIDs</a> : increased risk of kidney failure secondary to reduced kidney blood flow resulting from prostaglandin inhibition; monitor | Onset of diuresis: PO 30–60 min.<br>Duration: PO 3–4 h.<br>Half-life: 1.5 h.<br><br>Monitor electrolytes, urea, creatinine and daily weight.<br><br>Bumetanide 1 mg produces a diuretic response similar to furosemide 40 mg in healthy patients; in kidney failure, bumetanide 2 mg produces a diuretic response similar to furosemide 40 mg.<br><br>Contraindications: hepatic coma, states of severe electrolyte depletion. |

| Drug/Cost <a href="#">[a]</a>                                                                            | Dosage                                                                                                                                      | Adverse Effects                                                                                                                         | Drug Interactions                                                                                                                                     | Comments                                                                                                             |
|----------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|
|                                                                                                          |                                                                                                                                             |                                                                                                                                         | for reduced diuretic and natriuretic response.                                                                                                        |                                                                                                                      |
| <b>ethacrynic acid</b>  | Initial: 50 mg PO as single dose after a meal                                                                                               | Volume depletion.                                                                                                                       | Aminoglycosides, cisplatin: increased ototoxicity.                                                                                                    | May be considered in individuals with sulfonamide diuretic hypersensitivity or in cases of severe, refractory edema. |
| Edecrin                                                                                                  |                                                                                                                                             | Electrolyte disturbances: hypochloremia, hypokalemia (evident during first 1–2 wk), hyponatremia, hyperuricemia (usually asymptomatic). | Thiazide diuretics: increased diuresis and electrolyte abnormalities; monitor for dehydration and electrolyte abnormalities when used in combination. | Onset of diuresis: PO 30 min; IV 5 min.                                                                              |
| \$40–60                                                                                                  | Titrate gradually in increments of 25–50 mg daily until desired response; may require up to 200 mg BID in severe, refractory cases of edema | Azotemia, increase in serum creatinine.                                                                                                 | Digoxin: hypokalemia and hypomagnesemia may lead to arrhythmias.                                                                                      | Duration: PO 6–8 h; IV 2 h.                                                                                          |
|                                                                                                          | Maintenance: lowest effective daily dose                                                                                                    | Hyperglycemia.                                                                                                                          |                                                                                                                                                       | Half-life: 30–60 min.                                                                                                |
|                                                                                                          | When adding to another diuretic, start low (25 mg daily) and titrate as necessary                                                           | Dizziness, hypotension.                                                                                                                 | Lithium: reduced kidney clearance of lithium resulting in lithium toxicity; monitor lithium levels and adjust dose as needed.                         | Monitor electrolytes, urea, creatinine and daily weight.                                                             |
|                                                                                                          |                                                                                                                                             | Acute interstitial nephritis.                                                                                                           |                                                                                                                                                       | Similar in potency to furosemide.                                                                                    |
|                                                                                                          |                                                                                                                                             | Ototoxicity.                                                                                                                            |                                                                                                                                                       | IV route used when oral intake is not viable or in urgent situations such as pulmonary edema.                        |
|                                                                                                          |                                                                                                                                             | Myalgias (common with high doses of bumetanide).                                                                                        | <a href="#">NSAIDs</a> : increased risk of kidney failure secondary to reduced kidney blood flow                                                      |                                                                                                                      |
|                                                                                                          |                                                                                                                                             | Allergy (rare).                                                                                                                         | resulting from prostaglandin inhibition; monitor for reduced                                                                                          |                                                                                                                      |
|                                                                                                          |                                                                                                                                             | Ethacrynic acid may be                                                                                                                  |                                                                                                                                                       |                                                                                                                      |

| Drug/Cost <a href="#">[a]</a> | Dosage | Adverse Effects                                                         | Drug Interactions                  | Comments |
|-------------------------------|--------|-------------------------------------------------------------------------|------------------------------------|----------|
|                               |        | associated with a higher risk of ototoxicity than other loop diuretics. | diuretic and natriuretic response. |          |

| Drug/Cost <sup>[a]</sup>                                                                                                         | Dosage                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Adverse Effects                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Drug Interactions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Comments                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|----------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>furosemide</b> <br>Lasix, generics<br>< \$20 | Initial: 20–80 mg<br>PO as single<br>dose preferably<br>in a.m. If desired<br>response not<br>achieved,<br>increase by 20–<br>40 mg Q6–8H;<br>maximum single<br>dose dependent<br>on disease state<br>and kidney<br>function;<br>increase dose if<br>initial dose fails<br>to produce<br>adequate<br>increase in<br>urinary output<br>within 4–6 h<br><br>IV dose: 20–<br>40 mg as single<br>injection; may<br>repeat Q2H. If no<br>response, may<br>increase dose in<br>20 mg<br>increments<br><br>Continuous<br>infusion: 3–<br>20 mg/h titrated<br>to response<br><br>Maintenance:<br>lowest effective<br>dose taken daily,<br>BID or<br>intermittently on<br>alternate days | Volume<br>depletion.<br>Electrolyte<br>disturbances:<br>hypochloremia,<br>hypokalemia<br>(evident during<br>first 1–2 wk),<br>hyponatremia,<br>hyperuricemia<br>(usually<br>asymptomatic).<br><br>Azotemia,<br>increase in<br>serum<br>creatinine.<br><br>Hyperglycemia.<br><br>Dizziness,<br>hypotension.<br><br>Acute<br>interstitial<br>nephritis.<br><br>Ototoxicity.<br><br>Myalgias<br>(common with<br>high doses of<br>bumetanide).<br><br>Allergy (rare). | Aminoglycosides,<br>cisplatin: increased<br>ototoxicity.<br><br>Thiazide diuretics:<br>increased diuresis<br>and electrolyte<br>abnormalities;<br>monitor for<br>dehydration and<br>electrolyte<br>abnormalities when<br>used in<br>combination.<br><br>Digoxin:<br>hypokalemia and<br>hypomagnesemia<br>may lead to<br>arrhythmias.<br><br>Lithium: reduced<br>kidney clearance of<br>lithium resulting in<br>lithium toxicity;<br>monitor lithium<br>levels and adjust<br>dose as needed.<br><br><b>NSAIDs</b> : increased<br>risk of kidney<br>failure secondary<br>to reduced kidney<br>blood flow<br>resulting from<br>prostaglandin<br>inhibition; monitor<br>for reduced<br>diuretic and<br>natriuretic<br>response. | Onset of diuresis:<br>PO 30–60 min;<br>IV 5 min.<br><br>Duration: PO 4–8 h;<br>IV 2 h.<br><br>Half-life: 1.5–2 h,<br>prolonged in kidney<br>insufficiency.<br><br>Extent and rate of<br>absorption varies<br>between patients<br>depending on<br>concomitant<br>disease states.<br><br>Effectiveness<br>diminishes in the<br>presence of gut<br>wall edema.<br><br>Monitor<br>electrolytes, urea,<br>creatinine and daily<br>weight.<br><br>Furosemide 500 mg<br>tablet: for patients<br>with ClCr 5–<br>20 mL/min who<br>have not responded<br>to conventional<br>doses.<br><br>May use in<br>combination with<br>thiazides for<br>synergy. |

| Drug/Cost <sup>[a]</sup>                | Dosage                                                                                  | Adverse Effects | Drug Interactions | Comments |
|-----------------------------------------|-----------------------------------------------------------------------------------------|-----------------|-------------------|----------|
|                                         | Severe edema:<br>can use 250–<br>500 mg tablet PO<br>under direction<br>of a specialist |                 |                   |          |
| Drug Class: Potassium-Sparing Diuretics |                                                                                         |                 |                   |          |

| Drug/Cost <sup>[a]</sup>                                                                                                             | Dosage                                                                                                                                                                                                                                 | Adverse Effects                                                                                                                                                                                                                 | Drug Interactions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Comments                                                                 |
|--------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------|
| <b>amiloride</b> <br>Midamor<br>< \$20              | Initial: 5 mg daily PO; increase to 10 mg daily if necessary<br><br>If persistent hypokalemia, may increase to 15 mg, then 20 mg daily with careful monitoring of electrolytes<br><br>ClCr 10–50 mL/min: administer 50% of normal dose | Hyperkalemia—common if used as monotherapy; incidence greater in kidney impairment, diabetes and patients who are older. Headache, impotence.<br><br>Hyponatremia, hypochloremia when used in combination with other diuretics. | <a href="#">ACEI</a> and <a href="#">ARBs</a> : increased K <sup>+</sup> levels; monitor kidney function and potassium levels when used in combination, especially in patients with kidney impairment and type 2 diabetes mellitus; adjust dose as needed.<br>Potassium preparations: increased K <sup>+</sup> ; avoid combination unless documented hypokalemia; monitor K <sup>+</sup> levels.<br><br>Tacrolimus, cyclosporine: increased K <sup>+</sup> levels.<br><br><a href="#">NSAIDs</a> : may negate the diuretic effect and increase risk of <a href="#">NSAID</a> -induced kidney dysfunction. | Contraindications: significant kidney function impairment, hyperkalemia. |
| <b>eplerenone</b> <br>Inspra, generics<br>\$40–60 | Initial: 25 mg daily PO or every 2 days<br><br>Maximum: 50 mg daily PO                                                                                                                                                                 | Hyperkalemia, dehydration, dizziness, diarrhea, nausea.                                                                                                                                                                         | <a href="#">ACEI</a> and <a href="#">ARBs</a> : increased K <sup>+</sup> levels; monitor kidney function and potassium levels when used in combination,                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Avoid if ClCr <30 mL/min.                                                |

| Drug/Cost <sup>[a]</sup>                                                                                                     | Dosage                                                                                                                                                                                                                                       | Adverse Effects                              | Drug Interactions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Comments                                                             |
|------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|
|                                                                                                                              | If ClCr 30–50 mL/min, maximum dose is 25 mg daily<br><br>Continue initial dose for ≥7 days, then titrate accordingly<br><br>If adequate diuresis has not occurred, add a second diuretic that acts proximally in nephron for additive effect |                                              | especially in patients with kidney impairment and type 2 diabetes mellitus; adjust dose as needed.<br>Potassium preparations: increased K <sup>+</sup> ; avoid combination unless documented hypokalemia; monitor K <sup>+</sup> levels.<br><br>Tacrolimus, cyclosporine: increased K <sup>+</sup> levels.<br><br><a href="#">NSAIDs</a> : may negate the diuretic effect and increase risk of <a href="#">NSAID</a> -induced kidney dysfunction.<br><br>Avoid combining with strong inhibitors/inducers of CYP3A4, e.g., clarithromycin, carbamazepine, ketoconazole, phenytoin. |                                                                      |
| <b>finerenone</b> <br>Kerendia<br>\$120 | 10–20 mg PO daily                                                                                                                                                                                                                            | Hyperkalemia, hyperuricemia and hypotension. | Contraindicated in combination with strong inhibitors of CYP3A4, e.g., itraconazole.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Not recommended in patients with eGFR <25mL/min/1.73m <sup>2</sup> . |



| Drug/Cost <sup>[a]</sup>                                                                                                       | Dosage                                                                                                              | Adverse Effects                                                                                                                               | Drug Interactions                                                                                                                                                                                                                                          | Comments                                                                                                                                                                                       |
|--------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                                                                |                                                                                                                     |                                                                                                                                               | Avoid combination with inducers of CYP3A4, e.g., phenobarbital, rifampin.                                                                                                                                                                                  | Do not initiate if serum K <sup>+</sup> is >5 mmol/L. Measure serum K <sup>+</sup> 4 wk after initiating treatment, after dosage adjustments, and throughout treatment; adjust dose if needed. |
|                                                                                                                                |                                                                                                                     |                                                                                                                                               | Grapefruit juice may significantly increase the blood levels and effects of finerenone.                                                                                                                                                                    |                                                                                                                                                                                                |
|                                                                                                                                |                                                                                                                     |                                                                                                                                               | Increased hypotension in combination with furosemide.                                                                                                                                                                                                      |                                                                                                                                                                                                |
|                                                                                                                                |                                                                                                                     |                                                                                                                                               | Increased risk of hyperkalemia with <a href="#">ACEI</a> , <a href="#">ARBs</a> .                                                                                                                                                                          |                                                                                                                                                                                                |
| <b>spironolactone</b> <br>Aldactone, generics | Initial: 25 mg daily PO                                                                                             | Hyperkalemia—common if used as monotherapy; incidence greater in kidney impairment, diabetes and patients who are older. Headache, impotence. | <a href="#">ACEI</a> and <a href="#">ARBs</a> : increased K <sup>+</sup> levels; monitor kidney function and potassium levels when used in combination, especially in patients with kidney impairment and type 2 diabetes mellitus; adjust dose as needed. | Contraindications: significant kidney impairment, hyperkalemia.                                                                                                                                |
| < \$20                                                                                                                         | Increase by 25–50 mg Q5 days to a maximum of 200 mg daily                                                           |                                                                                                                                               |                                                                                                                                                                                                                                                            |                                                                                                                                                                                                |
|                                                                                                                                | If adequate diuresis has not occurred, add a second diuretic, which acts proximally in nephron for additive effects |                                                                                                                                               |                                                                                                                                                                                                                                                            |                                                                                                                                                                                                |
|                                                                                                                                | CICr 10–50 mL/min: administer Q12–24H                                                                               | Hyponatremia, hypochloremia when used in combination with other diuretics.                                                                    | Potassium preparations: increased K <sup>+</sup> ; avoid combination unless documented hypokalemia; monitor K <sup>+</sup> levels.                                                                                                                         |                                                                                                                                                                                                |

| Drug/Cost <sup>[a]</sup>                                                                                                | Dosage                                                | Adverse Effects                                  | Drug Interactions                                                                                                                                                                    | Comments                                                         |
|-------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------|--------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------|
|                                                                                                                         | CICr <10 mL/min: avoid                                | Gynecomastia.                                    | Tacrolimus, cyclosporine: increased K <sup>+</sup> levels.                                                                                                                           |                                                                  |
|                                                                                                                         |                                                       |                                                  | <a href="#">NSAIDs</a> : may negate the diuretic effect and increase risk of <a href="#">NSAID</a> -induced kidney dysfunction.                                                      |                                                                  |
|                                                                                                                         |                                                       |                                                  | Digoxin: increased plasma levels due to reduced kidney tubular secretion; spironolactone may interfere with digoxin radioimmunoassay, resulting in falsely increased digoxin levels. |                                                                  |
|                                                                                                                         |                                                       |                                                  | Salicylates: may result in sodium retention; monitor sodium levels and blood pressure.                                                                                               |                                                                  |
|                                                                                                                         |                                                       |                                                  | Mitotane: antagonism of mitotane activity; avoid combination.                                                                                                                        |                                                                  |
| Drug Class: Thiazide Diuretics                                                                                          |                                                       |                                                  |                                                                                                                                                                                      |                                                                  |
| <b>chlorthalidone</b> <br>generics | 50–100 mg once daily PO; can be given every other day | Usually well tolerated. Orthostatic hypotension. | Loop diuretics: increased diuresis and electrolyte abnormalities;                                                                                                                    | Onset of diuresis: 2–6 h. Duration: 24–72 h. Half-life: 24–55 h. |
| < \$20                                                                                                                  |                                                       |                                                  |                                                                                                                                                                                      |                                                                  |



| Drug/Cost <sup>[a]</sup>                                                                                                           | Dosage                                                                                                                                                                                                                            | Adverse Effects                                                                                                                                                                                                                                                                                                                                                                               | Drug Interactions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Comments                                                                                                                                                                                                                                                                                                                                         |
|------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">indapamide</a> <br>generics<br>< \$20 | Initial: 2.5 mg once daily PO; may increase to 5 mg daily if no response within 1 wk<br><br>When adding to a loop diuretic, start low and titrate as necessary<br><br>Ineffective as monotherapy in patients with ClCr <50 mL/min | Usually well tolerated.<br>Orthostatic hypotension.<br><br>Electrolyte abnormalities: hyperuricemia (usually asymptomatic), hyponatremia (common), hypochloremia, hypokalemia (evident during first 1–2 wk), hypercalcemia.<br><br>Headache, vertigo, dizziness, asthenia, muscle cramps.<br><br>Volume depletion.<br><br>Acute interstitial nephritis.<br><br>Thiazides can exacerbate gout. | Loop diuretics: increased diuresis and electrolyte abnormalities; monitor for dehydration and electrolyte abnormalities when used in combination.<br>Digoxin: hypokalemia and hypomagnesemia may lead to arrhythmias.<br><br>Antidiabetic medications: thiazides may increase blood sugar—seen at higher doses. Monitor blood glucose and adjust dose of antidiabetic medications as necessary.<br><br>Lithium: reduced kidney clearance of lithium resulting in toxicity; monitor lithium levels and adjust dose as needed.<br><br><a href="#">NSAIDs</a> : may negate the diuretic effect; increased risk of <a href="#">NSAID</a> - | Onset of diuresis: 1–3 h.<br>Duration: 8–12 h.<br>Half-life: 15–25 h.<br>Take with food or milk to decrease GI side effects.<br>Monitor electrolytes in patients who may be at increased risk of hypokalemia, e.g., patients on digoxin, patients with cardiac arrhythmias.<br>Contraindications: progressive and severe oliguria, hepatic coma. |

| Drug/Cost <sup>[a]</sup>                                                                                                                 | Dosage                                                                                                                                               | Adverse Effects                                                                                                                                                                                                                                                                                                                                                                               | Drug Interactions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Comments                                                                                                                                                                                                                                                                     |
|------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                                                                          |                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                               | induced kidney dysfunction.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                              |
| <a href="#">metolazone</a> <br>Zaroxolyn<br>< \$20–40 | Single daily dose is recommended; start low and titrate<br><br>2.5–20 mg daily PO<br><br>Ineffective as monotherapy in patients with ClCr <50 mL/min | Usually well tolerated.<br>Orthostatic hypotension.<br><br>Electrolyte abnormalities: hyperuricemia (usually asymptomatic), hyponatremia (common), hypochloremia, hypokalemia (evident during first 1–2 wk), hypercalcemia.<br><br>Headache, vertigo, dizziness, asthenia, muscle cramps.<br><br>Volume depletion.<br><br>Acute interstitial nephritis.<br><br>Thiazides can exacerbate gout. | Loop diuretics: increased diuresis and electrolyte abnormalities; monitor for dehydration and electrolyte abnormalities when used in combination.<br>Digoxin: hypokalemia and hypomagnesemia may lead to arrhythmias.<br><br>Antidiabetic medications: thiazides may increase blood sugar—seen at higher doses. Monitor blood glucose and adjust dose of antidiabetic medications as necessary.<br><br>Lithium: reduced kidney clearance of lithium resulting in toxicity; monitor lithium levels and adjust dose as needed.<br><br><a href="#">NSAIDs</a> : may negate the diuretic | Onset of diuresis: 1 h.<br>Duration: 12–24 h, depending on dose.<br>Half-life: 8–10 h.<br>May be given for a limited time period initially.<br>May be given indefinitely to patients who remain volume expanded, based on target weight.<br>Contraindications: hepatic coma. |

| Drug/Cost <sup>[a]</sup>                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Dosage | Adverse Effects | Drug Interactions                                                            | Comments |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|-----------------|------------------------------------------------------------------------------|----------|
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |        |                 | effect; increased risk of <a href="#">NSAID</a> -induced kidney dysfunction. |          |
| <p><sup>[a]</sup> Cost of 30-day supply; includes drug cost only.</p> <p> Dosage adjustment may be required in renal impairment; see <a href="#">Dosage Adjustment in Renal Impairment</a>.</p> <p><b>Abbreviations:</b> ACEI = angiotensin-converting enzyme inhibitor; ARB = angiotensin receptor blocker; CrCl = creatinine clearance; GI = gastrointestinal; NSAID = nonsteroidal anti-inflammatory drug</p> |        |                 |                                                                              |          |

## Suggested Readings

[Ellison DH. Clinical pharmacology in diuretic use. \*Clin J Am Soc Nephrol\* 2019;14\(8\):1248-57.](#)

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- [13. Soudet S, Lambert M, Lefèvre G et al. Long term use of metformin in idiopathic cyclic edema, report of thirteen cases and review of the literature. \*Pharmacol Res\* 2017;119:237-9.](#)

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